Leander F. Thiele

Kavli IPMU, 5-1-5 Kashiwanoha, Kashiwa City, Chiba 277-8583, Japan,
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EMPLOYMENT

Kavli IPMU, University of Tokyo — 2024–present
Project Assistant Professor (tenure-track) in Center for Data-Driven Discovery (CD3)

EDUCATION

Princeton University, Physics, PhD — 2019–2024
advisor: David N. Spergel
thesis: *Getting ready for new Data: Approaches to some Challenges in Cosmology*

Perimeter Institute, Theoretical Physics, MSc — 2018–2019
advisor: Kendrick M. Smith, co-advisor: J. Colin Hill
thesis: *Capturing non-Gaussianity: Analytic model for the one-point probability distribution function of cosmological fields within the halo model*

University of Oxford, Physics, BA First Class — 2015–2018
ranked top of the cohort ($\sim$ 130 students) in years 2 and 3

HONORS

Kusaka Memorial Prize in Physics (Princeton)	\$3000	2022
Member of the German Academic Scholarship Foundation	€35k	2015–2019
Perimeter Scholars International Award (Perimeter)	CA\$45k	2018
Scott Prize for best performance in the 3rd year (Oxford)	£400	2018
Winton Capital Prize for best performance in the 2nd year (Oxford)	£250	2017
BP Scholarship (Oxford)	£2000	2017
Rokos Award for summer research project (Oxford)	£800	2016

FUNDING

JSPS KAKENHI 26K17136 (early-career), *Reconstruction of the Local Universe with Machine Learning*	¥3.5M	2026
Royal Society International Collaboration Award Round 2 (Japan) 252140, with William Coulton, *Transformative joint analyses of transformative datasets*	£225k	2025
Google Research Ecosystem Grant for IPMU CD3	¥11M	2025
JSPS KAKENHI 24K22878 (research-activity startup), *AI staring into the void*	¥3M	2024

TEACHING

Astro-AI Asia Network (A3Net) summer school, Seoul, admin	2025
Lecture for UTokyo GUC, *Scientific Programming*	2025
Lecture & Tutorial for ILANCE students, IPMU, *Introduction to Machine Learning*	2025
Astro-AI Asia Network (A3Net) summer school, Osaka, *Basic Deep Learning*	2024

SERVICE

reviewer for ApJ, MNRAS, PRD, JCAP, NeurIPS ML4PS, ICML ML4Astro

ADVISING

Kristers Nagainis (Latvia/Leiden PhD)	assisted visitor	2026-
Di He (UTokyo PhD)	assisted	2025-
Lillie Szemraj (Princeton undergrad)	assisted visitor	2025
Felicia Xiao (MIT undergrad)	visitor	2025
Baptiste Barthe-Gold (Ecole Polytechnique Masters)	visitor	2025
Jessica Cowell (Oxford DPhil → Manchester PD)	assisted	2024-2025
Akira Tokiwa (UTokyo PhD → industry)	assisted	2024-2025
Cooper Jacobus (UC Berkeley undergrad → Stanford PhD)	visitor	2024
Bonny Wang (CMU Masters → UChicago PhD)	visitor	2024-

PUBLICATIONS

Citations: 1290, h-index: 20 (INSPIRE-HEP, as of 2026-04-02)

L. Thiele, *Bayesian Cosmic Void Finding with Graph Flows*, 2026,
`arXiv:2602.14630 [astro-ph.CO]`

J. Armijo, **L. Thiele**, J. Liu, *Replicating weak-lensing summary-statistic covariances with normalizing flows*, 2026,
`arXiv:2601.20669 [astro-ph.CO]`

G. Fabbian, F. Bianchini, A. Sabyr, J.C. Hill, C.C. Lovell, **L. Thiele**, D.N. Spergel, *A new constraint on the y-distortion with FIRAS: implications for feedback models in galaxy formation and cosmic shear measurements*, 2025,
`arXiv:2512.03038 [astro-ph.CO]`

A. Tokiwa, A.E. Bayer, J. Armijo, J. Liu, R. Terasawa, **L. Thiele**, M. Alvarez, L. Blot, M. Takada, *Impact of Simulation Box Size for Weak Lensing: Replication and Super-Sample Effects*, 2025,
`arXiv:2511.20423 [astro-ph.CO]`

B. Barthe-Gold, N.-M. Nguyen, **L. Thiele**, *Reconstructing the local density field with combined convolutional and point cloud architecture*, 2025,
`arXiv:2510.08573 [astro-ph.CO]`, poster at ML4PS workshop NeurIPS 2025

B.Y. Wang, **L. Thiele**, *Set-based Implicit Likelihood Inference of Galaxy Cluster Mass*, 2025,
`arXiv:2507.20378 [cs.LG]`, spotlight talk at ML4Astro workshop (colocated with ICML 2025)

J.A. Cowell, J. Armijo, **L. Thiele**, G.A. Marques, C.P. Novaes, D. Grandón, S. Cheng, M. Shirasaki, D. Alonso, J. Liu, *First Constraints from Marked Angular Power Spectra with Subaru Hyper Suprime-Cam Survey First-Year Data*, 2025,
`arXiv:2507.12315 [astro-ph.CO]`

L. Thiele, A.E. Bayer, N. Takeishi, *Simulation-Efficient Cosmological Inference with Multi-Fidelity SBI*, 2025,
`arXiv:2507.00514 [astro-ph.CO]`, poster at ML4Astro workshop (colocated with ICML 2025)

E. Calabrese, J.C. Hill, H.T. Jense, A. LaPosta, et al (incl. **L. Thiele**), *The Atacama Cosmology Telescope: DR6 Constraints on Extended Cosmological Models*, 2025,
JCAP 11, 063, `arXiv:2503.14454 [astro-ph.CO]`

C. Jacobus, **L. Thiele**, P. Harrington, J. Liu, Z. Lukic, *Enhancing Cosmological Simulations with Efficient and Interpretable Machine Learning in the Gabor Wavelet Basis*, 2024,
poster at Workshop on Machine Learning and the Physical Sciences (NeurIPS 2024)

L. Thiele, *De-baryonifying halos via optimal transport*, 2024,
`arXiv:2411.18399 [astro-ph.CO]`

J. Armijo, G.A. Marques, C.P. Novaes, **L. Thiele**, J.A. Cowell, D. Grandón, M. Shirasaki, J. Liu, *Cosmological constraints using Minkowski functionals from the first year data of the Hyper Suprime-Cam*, 2025,
MNRAS 537, 4, `arXiv:2410.00401 [astro-ph.CO]`

C.P. Novaes, **L. Thiele**, J. Armijo, S. Cheng, J.A. Cowell, G.A. Marques, E.G.M. Ferreira, M. Shirasaki, K. Osato, J. Liu, *Cosmology from HSC Y1 Weak Lensing with Combined Higher-Order Statistics and Simulation-based Inference*, 2025,
PRD 111, 8, `arXiv:2409.01301 [astro-ph.CO]`

A. Kogut, E. Switzer, D. Fixsen, N. Aghanim, J. Chluba, D. Chuss, J. Delabrouille, C. Dvorkin, B. Hensley, J.C. Hill, B. Maffei, A. Pullen, A. Rotti, A. Sabyr, **L. Thiele**, E. Wollack, I. Zelko, *The Primordial Inflation Explorer (PIXIE): Mission Design and Science Goals*, 2025,
JCAP 004, 020, `arXiv:2405.20403 [astro-ph.CO]`

S. Cheng, G.A. Marques, D. Grandón, **L. Thiele**, M. Shirasaki, B. Ménard, J. Liu, *Cosmological constraints from weak lensing scattering transform using HSC Y1 data*, 2025,
JCAP 01, 006, `arXiv:2404.16085 [astro-ph.CO]`

D. Grandón, G.A. Marques, **L. Thiele**, S. Cheng, M. Shirasaki, J. Liu, *Impact of baryonic feedback on HSC Y1 weak lensing non-Gaussian statistics*, 2024,
PRD 110, 103539, `arXiv:2403.03807 [astro-ph.CO]`

G.A. Marques, J. Liu, M. Shirasaki, **L. Thiele**, D. Grandón, K.M. Huffenberger, S. Cheng, J. Harnois-Déraps, K. Osato, W.R. Coulton, *Cosmology from weak lensing peaks and minima with Subaru Hyper Suprime-Cam survey first-year data*, 2023,
MNRAS 528, 3, `arXiv:2308.10866 [astro-ph.CO]`

L. Thiele, E. Massara, A. Pisani, C. Hahn, D.N. Spergel, S. Ho, B. Wandelt, *Neutrino mass constraint from an Implicit Likelihood Analysis of BOSS voids*, 2024,
ApJ 969, 89, `arXiv:2307.07555 [astro-ph.CO]`

L. Thiele, G.A. Marques, J. Liu, M. Shirasaki, *Cosmological constraints from HSC Y1 lensing convergence PDF*, 2023,
PRD 108, 123526, `arXiv:2304.05928 [astro-ph.CO]`

A.M. Delgado, D. Anglés-Alcázar, **L. Thiele**, M. Ntampaka, S. Pandey, K. Lehman, R.S. Somerville, S. Genel, F. Villaescusa-Navarro, *Predicting the impact of feedback on matter clustering with machine learning in CAMELS*, 2023,
MNRAS 526, 4, `arXiv:2301.02231 [astro-ph.GA]`

D. Wadekar, **L. Thiele**, J.C. Hill, S. Pandey, F. Villaescusa-Navarro, D.N. Spergel, M. Cranmer, D. Nagai, D. Anglés-Alcázar, S. Ho, L. Hernquist, *The SZ flux-mass (Y-M) relation at low halo masses: improvements with symbolic regression and strong constraints on baryonic feedback*, 2022,
MNRAS 522, 2, `arXiv:2209.02075 [astro-ph.CO]`

B.K.K. Lee, W. Coulton, **L. Thiele**, S. Ho, *An exploration of the properties of cluster profiles for the thermal and kinetic Sunyaev-Zel'dovich effects*, 2022,
MNRAS 517, 420, `arXiv:2205.01710 [astro-ph.CO]`

L. Thiele, M. Cranmer, W. Coulton, S. Ho, D.N. Spergel, *Predicting the Thermal Sunyaev-Zel'dovich Field using Modular and Equivariant Set-Based Neural Networks*, 2022,
MLST 3, 035002, `arXiv:2203.00026 [astro-ph.CO]`, poster at the Fourth Workshop on Machine Learning and the Physical Sciences (NeurIPS 2021)

L. Thiele, D. Wadekar, J.C. Hill, N. Battaglia, J. Chluba, F. Villaescusa-Navarro, L. Hernquist, M. Vogelsberger, D. Anglés-Alcázar, F. Marinacci, *Percent-level constraints on baryonic feedback with spectral distortion measurements*, 2022,
PRD 105, 083505, `arXiv:2201.01663 [astro-ph.CO]`

D. Wadekar, **L. Thiele**, F. Villaescusa-Navarro, J.C. Hill, D.N. Spergel, M. Cranmer, N. Battaglia, D. Anglés-Alcázar, L. Hernquist, S. Ho, *Augmenting astrophysical scaling relations with machine learning: application to reducing the SZ flux-mass scatter*, 2022,
PNAS 120(12), `arXiv:2201.01305 [astro-ph.CO]`

F. Villaescusa-Navarro, S. Genel, D. Anglés-Alcázar, L.A. Perez, P. Villanueva-Domingo, D. Wadekar, H. Shao, F.G. Mohammad, S. Hassan, E. Moser, E.T. Lau, L.F.M.P. Valle, A. Nicola, **L. Thiele**, Y. Jo, O.H.E. Philcox, B.D. Oppenheimer, M. Tillman, C. Hahn, N. Kaushal, A. Pisani, M. Gebhardt, A.M. Delgado, J. Caliendo, C. Kreisch, K.W.K. Wong, W.R. Coulton, M. Eickenberg, G. Parimbelli, Y. Ni, U.P. Steinwandel, V. La Torre, R. Dave, N. Battaglia, D. Nagai, D.N. Spergel, L. Hernquist, B. Burkhart, D. Narayanan, B. Wandelt, R.S. Somerville, G.L. Bryan, M. Viel, Y. Li, V. Irsic, K. Kraljic, M. Vogelsberger, *The CAMELS project: public data release*, 2022,
ApJS 265, 54, `arXiv:2201.01300 [astro-ph.CO]`

B. Maffei, M.H. Abitbol, N. Aghanim, J. Aumont, E. Battistelli, J. Chluba, X. Coulon, P. De Bernardis, M. Douspis, J. Grain, S. Gervasoni, J.C. Hill, A. Kogut, S. Masi, T. Matsumara, C. O Sullivan, L. Pagano, G. Pisano, M. Remazeilles, A. Ritacco, A. Rotti, V. Sauvage, G. Savini, S.L. Stever, A. Tartari, **L. Thiele**, N. Trappe, *BISOU: a balloon project to measure the CMB spectral distortions*, 2021,
16th Marcel Grossmann Meeting, `arXiv:2111.00246 [astro-ph.IM]`

F. Villaescusa-Navarro, S. Genel, D. Anglés-Alcázar, **L. Thiele**, R. Dave, D. Narayanan, A. Nicola, Y. Li, P. Villanueva-Domingo, B. Wandelt, D.N. Spergel, R.S. Somerville, J.M. Zorrilla Matilla, F.G. Mohammad, S. Hassan, H. Shao, D. Wadekar, M. Eickenberg, K.W.K. Wong, G. Contardo, Y. Jo, E. Moser, E.T. Lau, L.F.M.P. Valle, L.A. Perez, D. Nagai, N. Battaglia, M. Vogelsberger, *The CAMELS Multifield Dataset: Learning the Universe's Fundamental Parameters with Artificial Intelligence*, 2021,
ApJS 259, 61, `arXiv:2109.10915 [cs.LG]`

F. Villaescusa-Navarro, S. Genel, D. Anglés-Alcázar, D.N. Spergel, Y. Li, B. Wandelt, **L. Thiele**, A. Nicola, J.M. Zorilla Matilla, H. Shao, S. Hassan, D. Narayanan, R. Dave, M. Vogelsberger, *Robust marginalization of baryonic effects for cosmological inference at the field level*, 2021,
`arXiv:2109.10360 [astro-ph.CO]`

F. Villaescusa-Navarro, D. Anglés-Alcázar, S. Genel, D.N. Spergel, Y. Li, B. Wandelt, A. Nicola, **L. Thiele**, S. Hassan, J.M. Zorrilla Mattilla, D. Narayanan, R. Dave, M. Vogelsberger, *Multifield Cosmology with Artificial Intelligence*, 2021,
`arXiv:2109.09747 [astro-ph.CO]`

L. Thiele, Y. Guan, J.C. Hill, A. Kosowsky, D.N. Spergel, *Can small-scale baryon inhomogeneities resolve the Hubble tension? An investigation with ACT DR4*, 2021, PRD 104, 063535, `arXiv:2105.03003 [astro-ph.CO]`

L. Thiele, J.C. Hill, K.M. Smith, *Accurate Analytic Model for the Weak Lensing Convergence One-Point Probability Distribution Function and its Auto-Covariance*, 2020, PRD 102, 123545, `arXiv:2009.06547 [astro-ph.CO]`

L. Thiele, F. Villaescusa-Navarro, D.N. Spergel, D. Nelson, A. Pillepich, *Teaching neural networks to generate Fast Sunyaev Zel'dovich Maps*, 2020, ApJ 902, 129, `arXiv:2007.07267 [astro-ph.CO]`

R. Cayuso, O.J.C. Dias, F. Gray, D. Kubizňák, A. Margalit, J.E. Santos, R.G. Souza, **L. Thiele**, *Massive vector fields in Kerr–Newman and Kerr–Sen black hole spacetimes*, 2020,
JHEP 159, `arXiv:1912.08224 [hep-th]`

L. Thiele, C.A.J. Duncan, D. Alonso, *Disentangling magnification in combined shear clustering analyses*, 2020,
MNRAS 491, 1746, `arXiv:1907.13205 [astro-ph.CO]`

R. Cayuso, F. Gray, D. Kubizňák, A. Margalit, R.G. Souza, **L. Thiele**, *Principal Tensor Strikes Again: Separability of Vector Equations with Torsion*, 2019,
PLB 795, 650, `arXiv:1906.10072 [hep-th]`

L. Thiele, J.C. Hill, K.M. Smith, *Accurate analytic model for the thermal Sunyaev-Zel'dovich one-point probability distribution function*, 2019,
PRD 99, 103511, `arXiv:1812.05584 [astro-ph.CO]`

F. Dinc, M. Medvidovic, **L. Thiele**, *Effective Geometry Monte Carlo: A Fast and Reliable Simulation Framework for Molecular Communication*, 2019,
IEEE Access 7, 28635

F. Dinc, **L. Thiele**, B. C. Akdeniz, *The effective geometry Monte Carlo algorithm: applications to molecular communication*, 2019,
PLA 383, 2594, `arXiv:1809.06438 [cs.ET]`

TALKS

Fundamentals Seminar, IPMU		2/2026
AI4HEP East Asia, KEK Tsukuba	invited	1/2026
AI4KMI, Nagoya	invited	1/2026
3rd Shanghai Assembly		11/2025
28th IBIS, Naha	invited	11/2025
NPML conference, Tokyo	invited	10/2025
ML workshop @ McGill		9/2025
Mila, Montreal	invited	9/2025
21st Rencontres du Vietnam	invited	8/2025
AI for Fugaku seminar		7/2025
ML4Astro, UTokyo		5/2025
LeCosPA Meets IPMU, Taipei		4/2025
FOPM seminar, UTokyo		3/2025
Voids @ CPPM, Marseille	invited	2/2025
KICC, Cambridge UK		2/2025
FAIRS-Japan workshop Nagoya	invited	12/2024
11th KIAS workshop Gyeongju	invited	10/2024
Cosmo'24 Kyoto		10/2024

LSS Quest Osaka		6/2024
MPA cosmology seminar		5/2024
Yale cosmology seminar	invited	2/2024
AI4Phys @ IPMU		1/2024
CCA Cosmo x ML tristate meeting		10/2023
CMB constellation meeting KIPAC Stanford		10/2023
DESI lunch Berkeley Lab		10/2023
BCCP seminar UC Berkeley		10/2023
IPMU CD3 seminar		9/2023
Institute d'Astrophysique Spatiale Orsay		9/2023
Cosmo'23 Madrid		9/2023
Nagoya		4/2023
IPMU		3/2023
Princeton gravity group		2/2023
UCL Physics & Astronomy		9/2022
IAS astro coffee		3/2022
AAS 239		1/2022
Cosmology Talks		1/2022
Learn the Universe @ CCA		8/2021
CMB-S4 meeting		8/2021
MPA Garching		10/2020
German Astronomical Society		9/2020
Perimeter Institute		5/2020
Princeton/IAS cosmo lunch		5/2020
CCA Cosmo x ML		5/2020